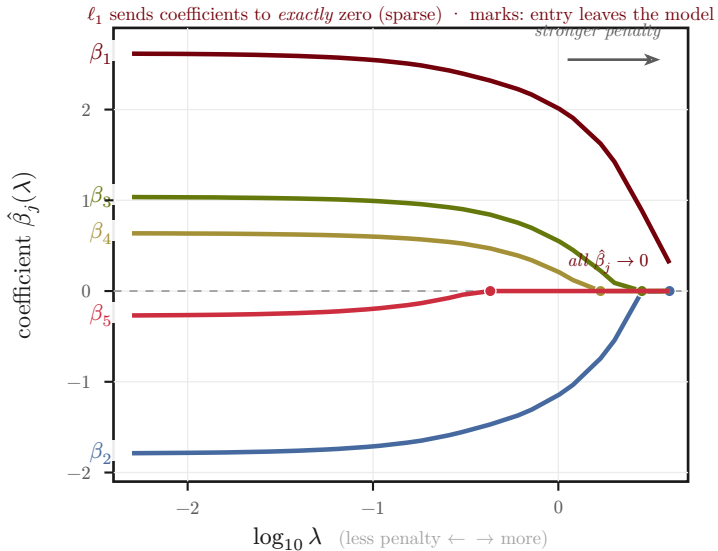


(a) Lasso regularization path

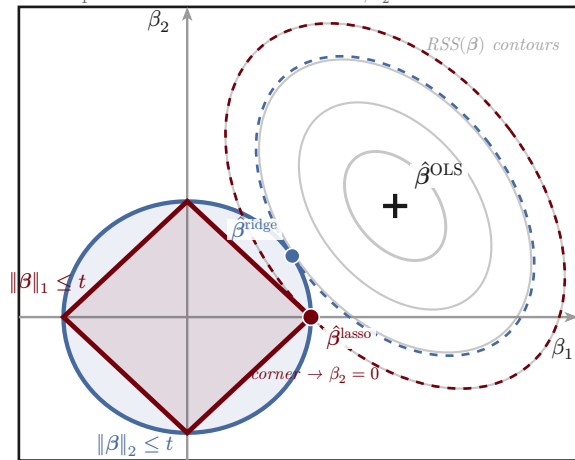
$$\hat{\beta}(\lambda) = \arg \min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \|\beta\|_1$$



(b) Constraint geometry

lowest RSS contour first touches the budget region

ℓ_1 diamond has corners on the axes; ℓ_2 circle is smooth



— ℓ_1 (lasso) $\rightarrow$ sparse — ℓ_2 (ridge) $\rightarrow$ shrink